PRUDHVI REDDY

Senior Agentic AI Data Engineer

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SUMMARY

AI Engineer with 6 years of software engineering experience and nearly 4 years building production AI, automation, and low code solutions on Microsoft Azure. Specializes in Python services that call Claude, OpenAI, and Azure OpenAI models, RAG and agent workflows, MCP servers that connect agents to enterprise tools and data, and Databricks Genie spaces that let business users query governed data in plain language. At Sam's Club, delivered AI agents used by about 2,000 merchandising and club operations users while cutting model spend by roughly 35%. Equally effective in code and in Copilot Studio and Power Platform, and known for shipping secure, well tested solutions that business teams adopt quickly.

CORE SKILLS

Languages:	Python, SQL, JavaScript
Generative AI and Agents:	Claude (Anthropic SDK), OpenAI API, Azure OpenAI, RAG, Prompt engineering, Tool calling, AI agents, Model Context Protocol (MCP) servers, LangChain, Semantic Kernel
Copilot and Power Platform:	Microsoft 365 Copilot, Copilot Studio, Power Automate, Power Apps, AI Builder, Custom connectors, Microsoft Teams
Data and Analytics:	Azure Databricks, Databricks AI/BI Genie, Databricks SQL, Unity Catalog, Apache Spark, Azure AI Search
APIs and Integration:	FastAPI, Flask, REST, JSON, Webhooks, ServiceNow APIs
Cloud:	Azure (Container Apps, Functions, App Service, Key Vault, Entra ID, Monitor)
Machine Learning:	PyTorch, TensorFlow, scikit-learn, NLP
Engineering Practices:	Git, GitHub Actions, Bicep, Docker, CI/CD, LLM evaluation, Tracing, Agile

PROFESSIONAL EXPERIENCE

Senior Agentic AI Engineer| Walmart (Client) | Bentonville, AR **Sep 2025 – Present**

- Led delivery of AI agents for Sam's Club merchandising and club operations, built in Python on Azure with Azure OpenAI and Claude models over Databricks data, answering inventory, sales, and supplier questions for about 2,000 users.
- Built MCP servers in Python that expose Databricks SQL warehouses, the item catalog, and ServiceNow as typed tools, letting agents in Copilot Studio and custom apps query live data under each user's permissions.
- Configured Databricks AI/BI Genie spaces over curated sales and inventory tables with instructions, SQL, and trusted assets, raising the correct answer rate on a 150 question benchmark from 68% to 91%.
- Developed a RAG service on Azure AI Search over merchandising policies, supplier agreements, and SOPs, using hybrid search, semantic reranking, and chunking tuned per document type to return cited answers.
- Engineered a FastAPI gateway that calls Claude, OpenAI, and Azure OpenAI through their Python SDKs with model routing, retries, prompt caching, and cost tracking, cutting monthly model spend by about 35%.
- Created Copilot Studio agents and Power Automate flows for club managers in Microsoft Teams, connected to the Python services through custom connectors and webhooks for approvals and notifications.
- Deployed services on Azure Container Apps and Azure Functions with Bicep and GitHub Actions, securing access with Key Vault, managed identities, and Entra ID.
- Evaluated prompts and agent workflows with automated test sets and tracing on accuracy, latency, and escalation rate, while the agents removed about 20 hours of manual weekly reporting per team.

AI Engineer| Los Angeles County (Client) | Los Angeles, CA **Mar 2024 – Sep 2025**

- Developed a Generative AI service agent in Copilot Studio for the Internal Services Department, backed by Python APIs on Azure, that answers IT and HR policy questions and resolves about 30% of routine requests without staff.
- Engineered a RAG pipeline in Python with Azure OpenAI embeddings and Azure AI Search over about 6,000 policy documents and knowledge articles, returning cited answers.
- Integrated the agent with ServiceNow and county HR systems through REST APIs and webhooks, so it checks ticket status and submits access requests on the user's behalf.
- Automated procurement request intake with Power Automate and AI Builder, extracting form fields and routing approvals, which cut request cycle time by about 40%.
- Built Azure Databricks pipelines in PySpark that clean and join service request, ticket, and HR data into curated Delta tables for reporting and agent context, governed through Unity Catalog permissions.

- Established prompt engineering standards with structured JSON outputs and few shot examples, testing each change against an evaluation set built with help desk staff.
- Enforced Azure AI Content Safety filters, PII redaction before model calls, and role-based access through Entra ID groups, keeping the agent aligned with county privacy and records policy.

AI Engineer | Mayo Clinic (Client) | **May 2023 – Dec 2023**

- Built an LLM chart summarization service in Python with Azure OpenAI that assembles EHR notes, labs, and visit history into structured prompts, cutting chart review from about 25 minutes to 8 per patient in a pilot with 40 care managers.
- Developed Flask REST APIs with JSON schemas, token-based authentication, and response caching that serve summaries and risk scores to care management applications, handling about 5,000 requests a day.
- Grounded every summary in care guidelines through a RAG pipeline with FAISS indexes and sentence transformer embeddings, chunking protocols by section so each summary cited the exact guideline it relied on.
- Designed prompt templates with structured JSON outputs and few shot clinical examples, versioning prompts in Git and comparing variants offline before each release.
- Validated outputs with nurse reviewers on 200 charts against agreed release thresholds, reducing hallucinated findings by about 60% across prompt and retrieval iterations.
- Protect patient data with deidentification before every model call, least privilege service accounts, and audit logging of each request, passing an internal HIPAA security review before go live.

Software Engineer| Dell Technologies (Client) | India **Jan 2019 to Dec 2021**

- Developed Python services and Flask REST APIs that expose device health, warranty status, and order history to support applications, handling about 50,000 requests a day at under 200 ms p95 latency.
- Built TensorFlow models that predict hardware component failures from device telemetry, scoring devices daily so support teams could open proactive cases before customers reported issues.
- Automated support case routing with an NLP classifier trained on historical case notes, assigning each case to the right product team and reducing manual triage time by about 30%.
- Created Power Automate flows with custom connectors to the warranty and order systems that pull device details into new support cases, saving agents about 4 minutes per case.
- Engineered Python and Spark pipelines on Azure HDInsight that clean and aggregate daily telemetry into device health features for model training and support reporting.
- Containerized services with Docker, built Jenkins CI pipelines with automated unit tests, and documented API contracts and integration runbooks for the operations team.

PROJECTS

Databricks MCP Server (open source) | Python, MCP, Databricks SQL, Claude

- Built an MCP server that exposes Databricks SQL warehouses and Unity Catalog metadata as tools, with a sample Claude agent that answers questions over governed tables.

EDUCATION

Master of Science (M.S.), Computer Science | University of Central Missouri | Jan 2022 – Dec 2023
Bachelor of Science (B.S.), Computer Science | Jawaharlal Nehru Technological University | Jun 2017 – Aug 2021

CERTIFICATIONS

Microsoft Certified Azure AI Engineer Associate (AI 102) | Microsoft Certified Power Platform Developer Associate (PL 400) | Databricks Certified Generative AI Engineer Associate